

Unit

1

Lesson

2

Spark The Challenge!

Design for Independence.

Time Frame: [90 min]

Lesson Objectives:

By the end of this lesson students will be able to:

- Understand and apply the Engineering Design Process (EDP).
- Develop empathy-driven design ideas for mobility support.
- Create a labeled sketch of a mobility device.

Challenge questions of the lesson

1. What support would help someone move more confidently?
2. How can design reflect empathy?

SEL Relation

- **Self-awareness:** Students reflect on their empathy maps (Engage) to identify how they connected with a real user's experience.
- **Social awareness:** Collaborative brainstorming based on diverse needs (Explore).
- **Responsible decision-making:** Evaluating which design best meets the user's needs (Think + Elaborate).
- **Relationship skills:** Peer-to-peer critique and team decision-making (Elaborate).
- **Self-management:** Planning and sketching an idea independently for the home challenge (Evaluate).

Engage



Teacher's Role:

- Facilitate a deep, reflective discussion around the core principles of design, iteration, and skill development.
- Guide students to synthesize information from all previous activities (DIY Lab, Builder, Appreciator, Video Lessons).
- Encourage critical thinking, diverse perspectives, and clear articulation of ideas.
- Summarize key takeaways to consolidate learning on the design process and necessary skills.

Expected From Students:

- Actively engage in critical thinking and self-reflection on the design process.
- Contribute thoughtfully to class discussions by sharing insights and connecting ideas.
- Articulate their understanding of how feedback drives iteration and the essential skills required for empathetic design.
- Demonstrate a comprehensive grasp of the iterative nature of the Engineering Design Process.

Parts of the book included.



Let's Think!



Objective:

Students will be able to critically reflect on the iterative nature of the design process, understand the crucial role of feedback, and identify the essential skills and mindsets required for creating effective and empathetic smart technology solutions for mobility challenges.

Answers:

How do you think Mechanical Engineers contribute in designing smart walkers?

Mechanical engineers play a critical role in designing smart walkers by focusing on the structural, functional, and ergonomic aspects of the device. They are responsible for:

1. **Designing the Framework:** Ensuring the walker is sturdy, lightweight, and easy to maneuver. Mechanical engineers work on the materials and structural integrity to make the walker safe and reliable.
2. **Incorporating Mobility Features:** They design mechanisms such as wheels, joints, and handles that enhance the walker's mobility, stability, and ease of use. They also consider how the walker adjusts to accommodate different users.
3. **Integrating Technology:** Mechanical engineers help integrate sensors, motors, and other smart technologies into the walker, enabling features like automatic braking, obstacle detection, or movement tracking to enhance user safety and comfort.
4. **User-Centered Design:** They also conduct research and gather data to understand the needs and limitations of users. This ensures the walker is accessible and functional for a diverse range of people, from elderly users to those with mobility impairments.
5. **Testing and Prototyping:** Engineers create prototypes and conduct tests to ensure the design meets safety standards and works effectively in real-life conditions.

Possible Strategies for implementation:

1. Think-Pair-Share:

- **Step 1 (Think):** Give students a few minutes to individually reflect on and write down their answers to each question.
- **Step 2 (Pair):** Have students discuss their answers with a partner, comparing their thoughts and expanding on each other's points.
- **Step 3 (Share):** Open up to a whole-class discussion, inviting pairs to share their synthesized insights.

2. Structured Discussion Prompts:

- Provide students with specific sentence starters or discussion prompts related to each question to guide their conversation.
- e.g., "One specific piece of feedback that drastically changed our design was...", "A skill I found most challenging to use was...", "I think the mindset of is most important because..."

3. Concept Mapping/Mind Mapping:

- Have students create a visual concept map or mind map individually or in small groups, connecting "Feedback," "Iteration," and "User Needs" or mapping out the "Skills for Design." This visually reinforces understanding.

Career Readiness



Objective:

Students will explore the role of biomedical engineers in improving movement, healing, and interaction with medical technology, connecting this to their understanding of design thinking and the Engineering Design Process.

Answers:

How do you think biomedical engineers can help people move, heal, or interact with medical technology?

- They design and build devices like prosthetic limbs, smart wheelchairs, and robotic exoskeletons that help people move more easily.
- They create medical equipment such as heart monitors, mobility aids, or rehabilitation machines that support healing.
- They develop smart technologies (like apps or sensors) that allow patients to control devices with their voice or movements.
- They improve the design of implants, wearable tech, or assistive tools that help people live more independently.
- They work with doctors and therapists to solve real problems patients face every day using technology.

Possible Strategies for implementation:

Strategy: "Biomedical Engineer's Impact Scan"

- **Description:** Students will actively explore different areas where biomedical engineers make a difference before synthesizing their understanding.
- **How to Implement:**
 - **Phase 1:** Initial Brainstorm & Question: Begin by posing the core question: "How do you think biomedical engineers help people move, heal, or interact with medical technology?" Students quickly jot down initial ideas individually.
 - **Phase 2:** Targeted Exploration (Video & Quick Research):
 1. Tell students: "We're going to explore some key areas where biomedical engineers work."
Play the video: <https://www.youtube.com/watch?v=ehlhTB1mymg>
 2. Optional: Provide 3-2 additional very short articles or images (e.g., a diagram of a prosthetic limb, a smart medical device, a tissue engineering concept) for groups to quickly scan after the video.
 - **Phase 3:** Group Synthesis & Share-Out: In small groups, have students discuss: "Based on the video and any additional resources, identify 2-3 specific ways biomedical engineers make a real-world impact related to mobility, healing, or medical tech." Invite groups to share their findings with the class.



Strategy: "Innovation Blueprint Challenge"

- **Description:** Students analyze a real-world biomedical innovation and consider the engineering process behind it.
- **How to Implement:**
 - **Phase 1:** Introduce Key Innovations: Present the class with 2-3 brief descriptions or images of real-world biomedical engineering innovations related to mobility or medical interaction (e.g., an advanced bionic prosthetic, a smart wearable health monitor, an exoskeleton for rehabilitation). You can use these video examples as starting points:
https://www.youtube.com/watch?v=izSBI0v9_Yk
 - **Phase 2:** Group Analysis: Divide students into small groups. Assign each group one of the innovations or allow them to choose.
 - **Phase 3:** Blueprint Questions: Provide each group with guiding questions for their chosen innovation:
 1. What specific problem does this innovation solve for users?
 2. What engineering skills (e.g., problem-solving, understanding of human body, technology integration) were likely crucial to designing this?
 3. Which steps of the Engineering Design Process do you think were most important in creating this device?
 4. What might have been some challenges or trade-offs in its design?



- **Phase 4: Class Showcase:** Have each group briefly present their chosen innovation and their analysis, highlighting the role of biomedical engineering in its creation.

Explain



Teacher's Role:

- Facilitate discussions and activities to help students understand the EDP.
- Guide students to apply the EDP to real-world scenarios.
- Encourage critical thinking, collaboration, and creative problem-solving.

Expected From Students:

- Actively participate in discussions and activities.
- Watch and analyze the provided video.
- Apply the Engineering Design Process (EDP) to a design challenge.
- Collaborate with peers to develop solutions.

Parts of the book included.



Screen Lessons

Now it's time to step into the role of engineers and use the Engineering Design Process (EDP) worksheet to find a solution to the problem

https://www.youtube.com/watch?v=MAhpfFt_mWM



Screen Lessons

Objective:

Students will understand the steps of the Engineering Design Process (EDP) and be able to apply it to a design challenge.

Video link: https://www.youtube.com/watch?v=MAhpfFt_mWM



Possible Strategies for implementation:

1. Pre-Watching: Setting the Stage

- **Strategy:** Introduce the Engineering Design Process (EDP) as a structured approach to problem-solving.
- **How to Implement:** Ask students: "Have you ever tried to solve a problem in a really organized way? What steps did you take?" Briefly discuss their experiences with problem-solving. Explain that the EDP is a systematic way that engineers approach challenges.

2. During Watching: Active Engagement with Key Concepts

- **Strategy:** Guide students to actively watch the video, focusing on identifying the steps of the EDP and relating them to the taco-making analogy.

- **How to Implement:** Ask students to take notes on each step of the EDP as its presented in the video. Encourage them to think about how each taco-making step relates to a similar step in solving an engineering problem.

3. Post-Watching: Connecting the Dots

- **Strategy:** Facilitate a discussion that connects the "Taco Party" analogy to real-world engineering challenges.
- **How to Implement:**
 - Ask: "How did the 'taco' video help you understand the Engineering Design Process? Can you give an example of a real-world problem that could be solved using these steps?"
 - Ask: "What are the key benefits of using the EDP when trying to solve a problem?"
 - Ask: "Which step of the EDP do you think is most important, and why?"

4. Hands-on Application: Design Challenge

- **Strategy:** Challenge students to apply the EDP to a specific design problem.
- **How to Implement:**
 - Present a design challenge, such as: "Design a device to help someone with limited mobility reach objects on a high shelf."
 - Divide students into groups and have them work through the steps of the EDP:
 1. **Define the Problem:** Clearly define the user's needs and the limitations of the situation.
 2. **Research:** What existing solutions are there? What materials are available?
 3. **Brainstorm:** Generate multiple ideas for a reaching device.
 4. **Prototype:** Create a simple model or sketch of their chosen design.
 5. **Test:** How would they test their design? What feedback would they seek?
 6. **Iterate:** How would they improve their design based on testing?
 7. **Share the Solution:** Present their final design and explain their process.

Engineering Design Process

Name:

Project:

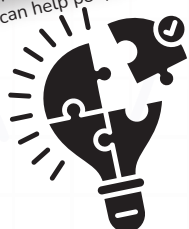
Ask

- What is the problem you are trying to solve?
Many people with spinal cord injuries or mobility challenges struggle to move independently, especially in outdoor or crowded environments. Traditional wheelchairs may require a lot of physical effort or assistance.
- What do users need?
A smart wheelchair that is easy to control, offers more independence, and works both indoors and outdoors.
- Why is this problem important to solve?
Because mobility is directly connected to freedom, confidence, and quality of life. Creating an easier, more accessible way to move can help people live more independently.



Imagine

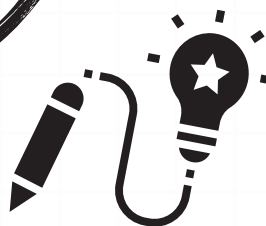
- Brainstorm different solutions:
A small prototype that moves with a joystick
A full-size version that includes sensors for better control
An upgrade with voice command features through a mobile app
Add safety features like automatic stop or alerts
- Choose the best solution to build first:
We will first design and build a joystick-controlled wheelchair on a small scale to test movement and turning. Then, we will scale it up and add a Bluetooth module and app interface to control it using voice commands.



Plan

What steps will you take?

- Build the small-scale model using motors, wheels, and a joystick.
- Test basic forward, backward, left, and right movement.
- Make adjustments for better stability and balance.
- Scale up the design using real-life wheelchair measurements.
- Add a Bluetooth module and program the mobile app using MIT App Inventor.
- Enable voice commands such as "forward," "stop," and "turn."
- Test the upgraded version for safety, response, and ease of use.



Create



Improve



Elaborate



Teacher's Role:

- Facilitate the design thinking process by guiding students through the challenge.
- Prompt students with critical thinking questions related to outdoor mobility and terrain challenges.
- Support brainstorming by providing examples of assistive technology and outdoor-use sensors.
- Encourage discussion around smart features, trade-offs, and user-centered design.
- Ensure students consider real-world contexts and inclusive design principles.

Expected From Students:

- Identify a realistic and meaningful mobility challenge specific to outdoor environments.
- Design a smart gadget that supports safe, independent navigation on uneven trails.
- Select appropriate sensors and actions (e.g., vibration alert, terrain-detecting sensors, GPS tracking).
- Analyse trade-offs in design decisions, such as size vs. stability or cost vs. performance.
- Begin sketching or modelling their proposed solution using available materials or tools.

Parts of the book included.



Builder

Objective:

Students will creatively apply their understanding of user empathy and smart technology to design a gadget that improves outdoor mobility for individuals like Alex. This activity

promotes innovative thinking, user-centered problem-solving, and a connection to inclusive design.

Activity Instructions:

1. In pairs or small teams, students will design a smart gadget to help a user named Alex, who uses a mobility aid, navigate uneven outdoor trails more safely and independently.
2. Students must complete all 5 design prompts:
 - Identify the user need
 - Describe their smart gadget idea
 - Specify the smart features (sensor, automatic action, communication)
 - List two design trade-offs
 - Suggest a refinement based on user feedback
3. Students may sketch, prototype, or describe their idea in writing using the worksheet or a digital tool (e.g., Google Slides, Jamboard).
4. Emphasize problem-solving, accessibility, and empathy in design decisions.
5. After completing their design, students can present it to peers for feedback.

Possible Strategies for Implementation:

- **Start with empathy mapping:** Have students first think about Alex's feelings, needs, and challenges when navigating a trail.
- **Show real examples:** Display images or videos of smart wheelchairs, terrain-detecting sensors, or assistive gadgets to inspire thinking.
- **Use design cards or prompts:** Provide cards with sensor types (e.g., ultrasonic, tilt, pressure), output ideas (LED, motor, buzzer), and mobility challenges to spark creativity.
- **Group roles:** Assign roles (researcher, sketcher, presenter, engineer) within each team to promote collaboration and equal participation.
- **Peer feedback round:** Allow students to share their initial designs with another group and gather quick feedback to refine their solution.
- **Display/share designs:** Encourage students to present their gadgets on a classroom wall or gallery walk with sticky note feedback.



Now I Can

Objective:

Students will reflect on and articulate their mastery of applying the Engineering Design Process to solve mobility challenges, recognizing the role of biomedical engineers, and understanding the importance of empathy in inclusive design.

Activity Instructions:

Step 1: Revisit the Learning Goals

Activity: Look back at the "Now I can" statements below. These represent the key skills and knowledge you've gained in this lesson about engineering for accessible futures.

Step 2: Student Self-Reflection

Activity: Individually, read each "Now I can" statement. Consider how well you understand and can perform that skill based on what you've learned and done in the lesson. Check the box next to each statement you feel confident in.

Step 3: Small Group Discussion

Activity: In your small groups, discuss each "Now I can" statement. Share specific examples from the activities (like the "Design for Outdoor Mobility" challenge or video discussions) that illustrate your understanding. Help each other clarify any points that are still unclear or provide further examples.

Relation to Standards

- **Engineering Design (HS-ETS1):** This entire lesson is a direct application of the Engineering Design Process.
 - **HS-ETS1-1:** Analyze a major global challenge to specify qualitative and quantitative criteria and constraints for solutions that account for societal needs and wants. (Applies to students defining Alex's specific challenge on uneven trails and identifying constraints for their gadget design.)
 - **HS-ETS1-2:** Design a solution to a complex real-world problem by breaking it down into smaller, more manageable problems that can be solved through engineering. (Applies as students brainstorm, sketch, and propose smart features for their gadget.)
 - **HS-ETS1-3:** Evaluate a solution to a complex real-world problem based on prioritized criteria and trade-offs that account for a range of constraints, including cost, safety, reliability, and aesthetics, as well as possible social, cultural, and environmental impacts. (Applies to students identifying and discussing design trade-offs for their gadget and considering user feedback for refinement.)

ISTE Standards for Students (Technology)

- **1.4 Innovative Designer:** Students use a variety of technologies within a design process to identify and solve problems by creating new, useful or imaginative solutions.
 - **1.4a:** Students know and use a deliberate design process for generating ideas, testing theories, creating innovative artifacts or solving authentic problems. (Directly applies to following the EDP to design the smart gadget.)
 - **1.4b:** Students select and use digital tools to plan and manage a design process that considers design constraints and calculated risks. (Applies if students use any digital sketching or planning tools; otherwise, it applies to the conceptual consideration of constraints.)
 - **1.4c:** Students develop, test and refine prototypes as part of a cyclical design process. (Applies to sketching prototypes and thinking about how user feedback would lead to refinement.)